# PulseIQ — Phase 3 Report: Money and insight

**Name:** Aali Harshad **Track:** Data Analytics **Phase:** 3 (Tasks 11–15) **Git tag:** `phase-3`
**Repository:** https://github.com/AALIHARSHAD/PluseIQ_Phase3_Project

## 1. Tasks and evidence

| Task | Done | Evidence |
|---|---|---|
| 11 Profit & loss, expense analytics | **Yes** | Page "Profit & loss" (Fig. 2). Group accounts excluded. Tie-out: 17 of 18 months equal the README to the paisa; Jun 2026 differs by ₹18,450 = planted DQ-04, explained on the page. |
| 12 Receivables and cash | **Yes** | Page "Receivables & cash" (Fig. 3). Buckets 22,133,626 + 3,036,258 + 1,940,982 + 112,283,412 = 139,394,278 = open receivables = README. Open statuses written on page. |
| 13 Cross-department view | **Yes** | Page "Department impact" (Fig. 4). 198 department-months × 13 measures match `reference_summary_dept_month`, 0 mismatches; raw-vs-clean gap explained (DQ-01/02/03). |
| 14 AI signals | **Yes** | Page "AI signals" (Fig. 5). Every chart labelled "AI-generated, for human review"; anomalies ranked by money at risk; forecast error table from back-test. |
| 15 Executive scorecard | **Yes** | Page "Company health scorecard" (Fig. 1) and `task15_scorecard_A4.pdf` (one A4 page). 8 KPIs, each from view `v_company_month`, each with a Details link. |

## 2. Proof it works

```
$ PYTHONPATH=src python -m pytest -q
23 passed, 1 skipped      (17 of them new in tests/test_phase3.py)
```

| Check | Mine | README / reference | Result |
|---|---|---|---|
| Revenue, 18 months | 1,260,930,052 | 1,260,930,052 | **✔ ties** |
| Net profit, 18 months | 224,665,137.58 | 224,683,587.58 | Δ 18,450 = DQ-04 (Jun 2026), explained |
| Open receivables (30 Sep 2026) | 139,394,278 | 139,394,278 | **✔ ties** |
| Bank balance, all 18 month-ends | rebuilt from ledger, max difference ₹0.00 | | ✔ |
| Department × month table | 198 rows | 198 rows | **✔ 0 mismatches** |
| My re-run of AI forecast method vs stored Oct 2026 | 426.28 | 426.3 | **✔ reproduces** |

**Forecast error (company headcount, my back-test):** MAPE 3.6% at 1 month ahead rising to 6.3% at 6 months; a naive "same as last month" guess is better at every horizon (1.3% → 5.8%); the 80% band caught the actual only 30–41% of the time, so it is too narrow. No stored forecast month has an actual yet (data ends where the forecast starts), so this is my back-test, not the AI team's own error.

## 3. What I learned

[Rewrite in your own words before submitting] Group accounts are headings and must never be added to their children. Profit is not cash: this company is profitable and still has ₹2.71 Cr overdue. Aggregate every source to the same grain first and join after, then re-add the totals. AI numbers need a range, an error figure and a label, and a simple baseline can beat the model. A scorecard needs a written rule for every colour.

## 4. Problems and how I solved them

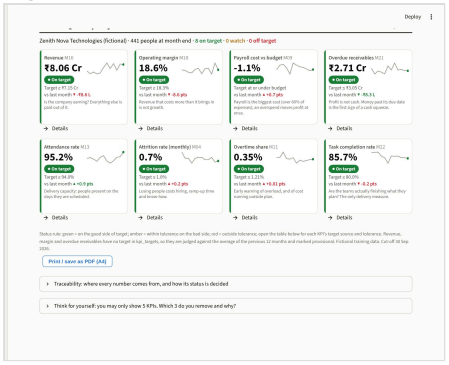
| | |
|---|---|
| Jun 2026 profit did not tie | Found ₹18,450 on one journal entry (JE-002094). Left in, flagged as DQ-04 instead of deleting to force a match. |
| Forecast has no actuals to test against | Re-ran the documented method from earlier months (back-test) and said so on the page. |
| AI page crashed (`.flags` column clashed with a pandas attribute) | Renamed to `n_flags`; page now runs; tests added. |
| kpi_targets has no thresholds, and no target for revenue, margin, overdue | Wrote the amber/red rule in `docs/decisions.md`; those three use the previous-12-month average and are marked provisional. **To confirm with finance.** |
| Scorecard printed on 2 pages | Print CSS removes sidebar and spacing; verified 1 A4 page. |

## 5. Plan for Phase 4

Package with Docker (`docker compose up phase3` already works), tag and release, add self-service filters, and keep the tie-out tests running in CI.

# Screenshots

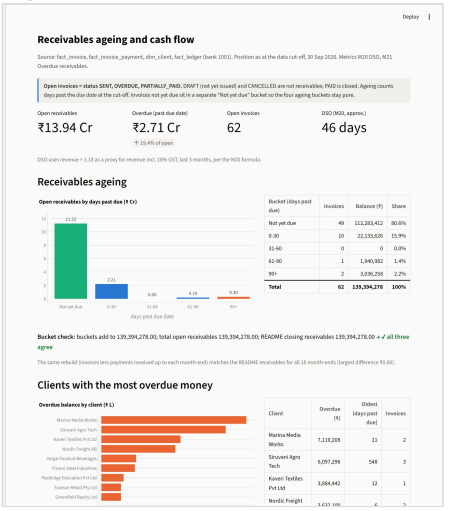
## Fig. 1 Scorecard



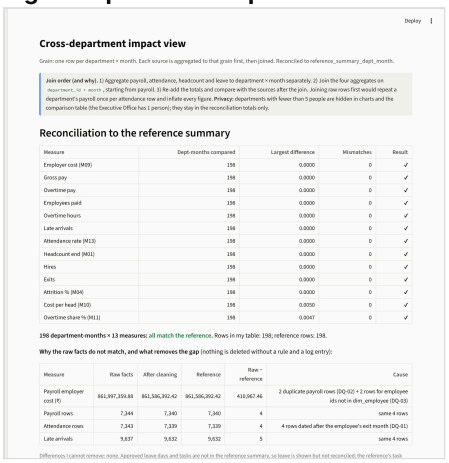

## Fig. 2 Profit & loss




## Fig. 3 Receivables & cash




## Fig. 4 Department impact




## Fig. 5 AI signals

